Q1.

A company repairs and sells computer hardware, including monitors, hard drives and keyboards.

Each monitor takes 3 hours to repair and the cost of components is £40.

Each hard drive takes 2 hours to repair and the cost of components is £20.

Each keyboard takes 1 hour to repair and the cost of components is £5.

Each month, the business has 360 hours available for repairs and £2500 available to buy components.

Each month, the company sells all of its repaired hardware to a local computer shop.

Each monitor, hard drive and keyboard sold gives the company a profit of £80, £35 and £15 respectively.

The company repairs and sells x monitors, y hard drives and z keyboards each month.

The company wishes to maximise its total profit.

(a) Find **five** inequalities involving x, y and z for the company's problem.

(3)

(b) (i) Find how many of each type of computer hardware the company should repair and sell each month.

(6)

(ii) Explain how you know that you had reached the optimal solution in part **(b)(i)**.

(1)

(iii) The local computer shop complains that they are not receiving one of the types of computer hardware that the company repairs and sells.

Using your answer to part **(b)(i)**, suggest a way in which the company's problem can be modified to address the complaint.

(1)

(Total 11 marks)

Q2.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = x - 2y + 3z$

subject to $x + y + z \leq 16$

$x - 2y + 2z \leq 17$

$$2x - y + 2z \leq 19$$

and $x \geq 0$, $y \geq 0$, $z \geq 0$.

(2)

(b) (i) The first pivot to be chosen is from the z-column. Identify the pivot and explain why this particular value is chosen.

(2)

(ii) Perform one iteration of the Simplex method.

(3)

(c) (i) Perform one further iteration.

(3)

(ii) Interpret the tableau that you obtained in part (c)(i) and state the values of your slack variables.

(3)

(Total 13 marks)

Q3.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = 4x + 3y + z$

subject to

$$2x + y + z \leq 25$$

$$x + 2y + z \leq 40$$

$$x + y + 2z \leq 30$$

and $x \geq 0$, $y \geq 0$, $z \geq 0$,

(2)

(b) The first pivot to be chosen is from the x-column.

Perform one iteration of the Simplex method.

(3)

(c) (i) Perform one further iteration.

(3)

(ii) Interpret your final tableau and state the values of your slack variables.

(3)

(Total 11 marks)

Q4.

A linear programming problem consists of maximising an objective function P involving three variables, x, y and z, subject to constraints given by three inequalities other than $x \geq 0$, $y \geq 0$ and $z \geq 0$. Slack variables s, t and u are introduced and the Simplex method is used to solve the problem. One iteration of the method leads to the following tableau.

P	x	y	z	s	t	u	***value***
1	−2	11	0	3	0	0	6
0	2	3	1	1	0	0	2
0	6	−30	0	−6	1	0	3
0	−1	−9	0	−3	0	1	4

(a) (i) State the column from which the pivot for the **next** iteration should be chosen. Identify this pivot and explain the reason for your choice.

(3)

(ii) Perform the next iteration of the Simplex method.

(4)

(b) (i) Explain why you know that the maximum value of P has been achieved.

(1)

(ii) State how many of the three original inequalities still have slack.

(1)

(c) (i) State the maximum value of P and the values of x, y and z that produce this maximum value.

(2)

(ii) The objective function for this problem is $P = kx - 2y + 3z$, where k is a constant.

Find the value of k.

(2)

(Total 13 marks)

Q5.

(a) Given that k is a constant, complete the Simplex tableau below for the following linear programming problem.

Maximise $P = kx + 6y + 5z$

subject to

$$2x + y + 4z \leq 11$$
$$x + 3y + 6z \leq 18$$
$$x \geq 0,\ y \geq 0,\ z \geq 0$$

P	x	y	z	s	t	***value***

1	$-k$	−6	−5	0	0	0
0						
0						

(2)

(b) Use the Simplex method to perform **one** iteration of your tableau for part (a), choosing a value in the ***y*-column** as pivot.

P	x	y	z	s	t	*value*

(4)

(c) (i) In the case when $k = 1$, explain why the maximum value of P has now been reached and write down this maximum value of P.

(2)

(ii) In the case when $k = 3$, perform one further iteration and interpret your new tableau.

P	x	y	z	s	t	*value*

(6)
(Total 14 marks)

Q6.

The Simplex method is to be used to maximise $P = 3x + 2y + z$ subject to the constraints

$$-x + y + z \leq 4$$

$$2x + y + 4z \leq 10$$

$$4x + 2y + 3z \leq 21$$

The initial Simplex tableau is given below.

P	x	y	z	s	t	u	*value*
1	−3	−2	−1	0	0	0	0

0	−1	1	1	1	0	0	4
0	2	1	4	0	1	0	10
0	4	2	3	0	0	1	21

(a) (i) The first pivot is to be chosen from the x-column. Identify the pivot and explain why this particular value is chosen.

(2)

(ii) Perform one iteration of the Simplex method and explain how you know that the optimal value has not been reached.

(5)

(b) (i) Perform one further iteration.

(4)

(ii) Interpret the final tableau and write down the initial inequality that still has slack.

(4)

(Total 15 marks)

Q7.

A linear programming problem involving variables x, y and z is to be solved. The objective function to be maximised is $P = 2x + 6y + kz$, where k is a constant.

The initial Simplex tableau is given below.

P	x	y	z	s	t	u	***value***
1	−2	−6	$-k$	0	0	0	0
0	5	3	10	1	0	0	15
0	7	6	4	0	1	0	28
0	4	3	6	0	0	1	12

(a) In addition to $x \geq 0$, $y \geq 0$, $z \geq 0$, write down **three** inequalities involving x, y and z for this problem.

(2)

(b) (i) By choosing the first pivot **from the y-column**, perform **one** iteration of the Simplex method.

(4)

(ii) Given that the optimal value has **not** been reached, find the possible values of k.

(2)

(c) In the case when $k = 20$:

(i) perform one further iteration;

(4)

(ii) interpret the final tableau and state the values of the slack variables.

(3)
(Total 15 marks)

Q8.

A linear programming problem involving variables x, y and z is to be solved. The objective function to be maximised is $P = 2x + 4y + 3z$. The initial Simplex tableau is given below.

P	x	y	z	s	t	u	value
1	–2	–4	–3	0	0	0	0
0	2	2	1	1	0	0	14
0	–1	1	2	0	1	0	6
0	4	4	3	0	0	1	29

(a) (i) What name is given to the variables s, t and u?

(1)

(ii) Write down an equation involving x, y, z and s for this problem.

(1)

(b) (i) By choosing the first pivot **from the y-column**, perform **one** iteration of the Simplex method.

(4)

(ii) Explain how you know that the optimal value has not been reached.

(1)

(c) (i) Perform one further iteration.

(4)

(ii) Interpret the final tableau, stating the values of P, x, y and z.

(3)
(Total 14 marks)

Q9.

(a) Given that k is a constant, display the following linear programming problem in a Simplex tableau.

Maximise $P = 6x + 5y + 3z$

subject to
$$x + 2y + kz \le 8$$
$$2x + 10y + z \le 17$$
$$x \ge 0,\ y \ge 0,\ z \ge 0$$

(3)

(b) (i) Use the Simplex method to perform **one** iteration of your tableau for part (a), choosing a value in the x-column as pivot.

(4)

(ii) Given that the maximum value of P has not been achieved after this first iteration, find the range of possible values of k.

(2)

(c) In the case where $k = -1$, perform one further iteration and interpret your final tableau.

(6)

(Total 15 marks)

Q10.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = 4x - 5y + 6z$

subject to $6x + 7y - 4z \leq 30$

$2x + 4y - 5z \leq 8$

$x \geq 0, y \geq 0, z \geq 0$

(3)

(b) The Simplex method is to be used to solve this problem.

(i) Explain why it is not possible to choose a pivot from the z-column initially.

(1)

(ii) Identify the initial pivot and explain why this particular element should be chosen.

(2)

(iii) Perform one iteration using your initial tableau from part (a).

(3)

(iv) State the values of x, y and z after this first iteration.

(2)

(v) Without performing any further iterations, explain why P has no finite maximum value.

(1)

(c) Using the same inequalities as in part (a), the problem is modified to

Maximise $Q = 4x - 5y - 20z$

(i) Write down a modified initial tableau and the revised tableau after one iteration.

(2)

(ii) Hence find the maximum value of Q.

(1)
(Total 15 marks)

Q11.

A linear programming problem involving variables x, y and z is to be solved. The objective function to be maximised is $P = 4x + y + kz$, where k is a constant. The initial Simplex tableau is given below.

P	x	y	z	s	t	*value*
1	–4	–1	$-k$	0	0	0
0	1	2	3	1	0	7
0	2	1	4	0	1	10

(a) In addition to $x \geq 0$, $y \geq 0$ and $z \geq 0$, write down **two** inequalities involving x, y and z for this problem.

(1)

(b) (i) The first pivot is chosen from the ***x*-column**. Identify the pivot and perform **one** iteration of the Simplex method.

(4)

(ii) Given that the optimal value of P has not been reached after this first iteration, find the possible values of k.

(2)

(c) Given that $k = 10$:

(i) perform one further iteration of the Simplex method;

(4)

(ii) interpret the final tableau.

(3)
(Total 14 marks)

Q12.

A linear programming problem involving the variables x, y and z is to be solved. The objective function to be maximised is $P = 2x + 3y + 5z$. The initial Simplex tableau is given below.

P	x	y	z	s	t	u	*value*
1	– 2	– 3	– 5	0	0	0	0
0	1	0	1	1	0	0	9

0	2	1	4	0	1	0	40
0	4	2	3	0	0	1	33

(a) In addition to $x \geq 0$, $y \geq 0$, $z \geq 0$, write down **three** inequalities involving x, y and z for this problem.

(2)

(b) (i) By choosing the first pivot from the z-column, perform **one** iteration of the Simplex method.

(4)

(ii) Explain how you know that the optimal value has not been reached.

(1)

(c) (i) Perform one further iteration.

(4)

(ii) Interpret the final tableau and state the values of the slack variables.

(3)

(Total 14 marks)

Q13.

A linear programming problem consists of maximising an objective function P involving three variables x, y and z. Slack variables s, t, u and v are introduced and the Simplex method is used to solve the problem. Several iterations of the method lead to the following tableau.

P	x	y	z	s	t	u	v	*value*
1	0	–12	0	5	–3	0	0	37
0	1	–8	0	1	2	0	0	16
0	0	4	0	0	3	0	1	20
0	0	2	0	–3	2	1	0	14
0	0	1	1	2	5	0	0	8

(a) (i) The pivot for the next iteration is chosen from the y-column. State which value should be chosen and explain the reason for your choice.

(2)

(ii) Perform the next iteration of the Simplex method.

(4)

(b) Explain why your new tableau solves the original problem.

(1)

(c) State the maximum value of P and the values of x, y and z that produce this maximum value.

(2)

(d) State the values of the slack variables at the optimum point. Hence determine how many of the original inequalities still have some slack when the optimum is reached.

(2)
(Total 11 marks)

Q14.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = 5x + 8y + 7z$

subject to
$$3x + 2y + z \leq 12$$
$$2x + 4y + 5z \leq 16$$
$$x \geq 0,\ y \geq 0,\ z \geq 0$$

(3)

(b) The Simplex method is to be used by initially choosing a value in the y-column as a pivot.

(i) Explain why the initial pivot is 4.

(1)

(ii) Perform **two** iterations of your tableau from part (a) using the Simplex method.

(6)

(iii) State the values of P, x, y and z after your second iteration.

(2)

(iv) State, giving a reason, whether the maximum value of P has been achieved.

(1)
(Total 13 marks)

Q15.

A linear programming problem involving variables x and y is to be solved. The objective function to be maximised is $P = 3x + 5y$. The initial Simplex tableau is given below.

P	x	y	s	t	u	*value*
1	–3	–5	0	0	0	0
0	1	2	1	0	0	36
0	1	1	0	1	0	20
0	4	1	0	0	1	39

(a) In addition to $x \geq 0$, $y \geq 0$, write down **three** inequalities involving x and y for this problem.

(2)

(b) (i) By choosing the first pivot from the ***y*-column**, perform **one** iteration of the Simplex method.

(4)

(ii) Explain how you know that the optimal value has not been reached.

(1)

(c) (i) Perform one further iteration.

(4)

(ii) Interpret the final tableau and state the values of the slack variables.

(3)

(Total 14 marks)

Q16.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = 3x + 2y + 4z$

subject to

$$x + 4y + 2z \leq 8$$
$$2x + 7y + 3z \leq 21$$
$$x \geq 0,\ y \geq 0,\ z \geq 0$$

(3)

(b) Use the Simplex method to perform **one** iteration of your tableau for part (a), choosing a value in the z-column as pivot.

(3)

(c) (i) Perform one further iteration.

(5)

(ii) State whether or not this is the optimal solution, and give a reason for your answer.

(2)

(Total 13 marks)

Q17.

A linear programming problem involving variables x and y is to be solved. The objective function to be maximised is $P = 4x + 9y$. The initial Simplex tableau is given below.

P	x	y	r	s	t	value
1	–4	–9	0	0	0	0
0	3	7	1	0	0	33
0	1	2	0	1	0	10
0	2	7	0	0	1	26

(a) Write down the **three** inequalities in x and y represented by this tableau.

(2)

(b) The Simplex method is to be used to solve this linear programming problem by initially choosing a value in the x-column as the pivot.

(i) Explain why the initial pivot has value 1.

(2)

(ii) Perform **two** iterations using the Simplex method.

(7)

(iii) Comment on how you know that the optimum solution has been achieved and state your final values of P, x and y.

(3)

(Total 14 marks)

Q18.

A linear programming problem involving variables x and y is to be solved. The objective function to be maximised is $P = 3x + 2y$. The initial Simplex tableau is given.

p	x	y	r	s	t	*value*
1	–3	–2	0	0	0	0
0	4	5	1	0	0	36
0	2	1	0	1	0	12
0	5	2	0	0	1	35

(a) In addition to $x \geq 0$, $y \geq 0$, write down **three** inequalities in this problem.

(2)

(b) (i) Perform two iterations using the Simplex method, by initially choosing a value in the x-column as a pivot, to solve this linear programming problem. Comment on how you know that the optimum solution has been achieved.

(7)

(ii) State your final values of P, x and y.

(2)

(iii) State the values of the slack variables r, s and t at the optimum solution.

(1)

(Total 12 marks)

Q19.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = 2x + 3y$

subject to $2x + y \leq 20$
$x + 2y \leq 20$
$5x + 4y \leq 60$

(2)

(b) Solve the problem using the Simplex algorithm, giving your answers as exact fractions.

(9)
(Total 11 marks)

Q20.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise

$$P = 2x - 3y + z$$

subject to

$$3x + 6y + z \leq 72$$
$$4x + 2y + z \leq 48$$
$$x - y + z \leq 36.$$

(2)

(b) Solve the problem using the Simplex algorithm, stating the values of P, x, y, z and the slack variables.

(10)
(Total 12 marks)

Q21.

(a) A student is solving a linear programming question using the Simplex algorithm. The student obtains the following Simplex tableau.

P	x	y	z	r	s	
5	0	0	8	1	9	12
0	5	0	–2	1	–1	2
0	0	15	12	–1	6	3

(i) Explain why this tableau is optimal.

(1)

(ii) Solve this tableau, stating the values of x, y, z and P.

(2)

(b) Solve the following linear programming problem using the Simplex algorithm.

Maximise $P = 3x + 6y + 2z,$

subject to $3x + 4y + 2z \leq 2,$
$x + 3y + 2z \leq 1$
and $x \geq 0, \quad y \geq 0, \quad z \geq 0$

(9)
(Total 12 marks)

Q22.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $3x + 6y + 2z$

subject to $x + 3y + 2z \leq 11$

$3x + 4y + 2z \leq 21$

and $x \geq 0, \quad y \geq 0, z \geq 0.$

(2)

(b) Solve the problem using the Simplex algorithm.

(8)
(Total 10 marks)

Q23.

Asif and Bharveen play a zero-sum game. The game is represented by the following pay-off matrix for Asif.

		Bharveen		
		I	**II**	**III**
Asif	**I**	4	2	6
	II	2	10	1
	III	3	1	5

(a) Explain why Asif should never play strategy **III**.

(1)

(b) Asif plays strategy **I** with probability p_1 and strategy **II** with probability p_2. The value of the game is v. Use the matrix to write down **three** inequalities in the form $v \leq ap_1 + bp_2$.

(3)

The probabilities p_1 and p_2 also satisfy the inequalities

$$p_1 + p_2 \leq 1, \qquad p_1 \geq 0 \qquad \text{and} \qquad p_2 \geq 0.$$

The Simplex tableau for the linear programming problem which will give the optimal strategy for Asif can be written as:

P	v	p_1	p_2	r	s	t	u	
1	–1	0	0	0	0	0	0	0
0	0	1	1	1	0	0	0	1
0	1	–4	–2	0	1	0	0	0
0	1	–2	–10	0	0	1	0	0
0	1	–6	–1	0	0	0	1	0

(c) (i) Apply **one** iteration to this Simplex tableau.

(4)

(ii) Explain why your answer to part (c)(i) is **not** optimal.

(1)

(iii) After two further iterations, Asif s tableau is:

P	v	p_1	p_2	r	s	t	u	
5	0	0	0	18	4	1	0	18
0	0	0	10	2	1	–1	0	2
0	0	10	0	8	–1	1	0	8
0	5	0	0	18	4	1	0	18
0	0	0	0	14	–13	3	10	14

Show that the value of the game in this case is $3\frac{3}{5}$ and find Asif's optimal mixed strategy.

(3)

(d) Assuming the value of the game is $3\frac{3}{5}$ and by considering mixed strategies, find the optimal mixed strategy for Bharveen.

(6)
(Total 18 marks)

Q24.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = 8x + 7y$

subject to $2x + y \leq 32$

and $x + y \leq 20$

(2)

(b) Solve the problem using the Simplex algorithm.

(7)
(Total 9 marks)

Q25.

A linear programming problem in x and y is to be solved. Part of the initial tableau is given below.

x	y	r	s	t	
4	3	1	0	0	33
–1	1	0	1	0	4
2	5	0	0	1	27

(a) In addition to $x \geq 0$ and $y \geq 0$, write down the **three** inequalities in this problem.

(2)

(b) (i) The objective function $P = 2x + 2y$ is to be maximised.

Solve this linear programming problem using the simplex algorithm, by initially using a value in the x column as the pivot. (You do **not** require more than two iterations.)

(7)

(ii) State your final values of P, x and y.

(2)
(Total 11 marks)

Q26.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = 4x + 5y + 3z$

subject to $8x + 5y + 2z \leq 3$

$4x + 6y + 9z \leq 2$

and $x \geq 0,\ y \geq 0,\ z \geq 0$

(2)

(b) Solve the problem using the Simplex algorithm, giving your answers as exact fractions.

(9)
(Total 11 marks)

Q27.

(a) Display the following linear programming problem in a Simplex tableau.

Maximise $P = 3x + 5y + 4z$,

subject to $3x + 2y + z \leq 24$,

$2x + y + 2z \leq 20,$

and $x \geq 0, y \geq 0, z \geq 0.$

(2)

(b) Solve the problem using the Simplex algorithm, giving your answers as exact fractions.

(9)

(Total 11 marks)

Q28.

A linear programming problem consists of maximising an objective function P in three variables x, y and z. The simplex method is used to solve the problem and several iterations of the method lead to the following tableau:

P	x	y	z	s	t	u	v	
1	–2	0	0	2	0	0	1	60
0	–2	0	1	1	0	0	–1	15
0	1	0	0	2	1	0	2	25
0	2	0	0	–1	0	1	1	40
0	1	1	0	0	0	0	4	30

(a) What is the name given to the variables s, t, u and v?

(1)

(b) Other than $x \geq 0$, $y \geq 0$ and $z \geq 0$, how many inequalities are involved in the problem?

(1)

(c) Apply one further iteration of the simplex method to the above tableau.

(5)

(d) Explain why your new tableau solves the original problem. State the maximum value of P and the values of x, y and z for which that maximum is reached.

(3)

(e) State the values of s, t, u and v at the optimal point. How many of the inequalities from the original problem still have some slack?

(2)

(Total 12 marks)

Q29.

A drug company produces three types of popular painkiller: Xtrafast, Yourelax and Zizz. Each of these is a combination of three key ingredients: aspirin, bupro and codeine. The table below shows the numbers of units of each of the three ingredients required in making batches of the three painkillers. It also shows the amounts of the ingredients available and the profit on each batch of painkillers.

	aspirin	bupro	codeine	profit
Xtrafast	1	1	3	£15
Yourelax	2	1	6	£10
Zizz	1	1	2	£10
Amount available	60	55	140	

The company wishes to maximise its profits.

Let x be the number of batches of Xtrafast made, y the number of Yourelax and z the number of Zizz.

(a) State the problem as a linear programming problem, writing down the objective function and the full set of inequalities.

(3)

(b) Copy and complete the following initial tableau for the simplex method when applied to this problem:

P	x	y	z	s	t	u	
1	–15	–10	–10	0	0	0	0
0	1	2	1	1	0	0	60
.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.

(2)

(c) Perform one iteration of the simplex method by increasing z.

(5)

(d) Perform a second iteration of the simplex method.

(4)

(e) State the maximum profit possible and how many batches of Xtrafast, Yourelax and Zizz should be made in order to obtain that profit.

Explain why some people might not be happy with this solution.

(3)

(Total 17 marks)

Q30.

A furniture company can produce stools, armchairs and settees, each of which requires components A, B and C. The table below shows the numbers of components needed, the numbers of components available, and the profit on each item of furniture.

	Component A	Component B	Component C	Profit per item
Stool	2	1	2	£20
Armchair	1	1	3	£10
Settee	2	1	3	£30
Number available	110	60	140	

he company wishes to maximise its profits. Let x be the number of stools made, y the number of armchairs made, and z the number of settees made.

(a) State the problem as a linear programming problem, writing down the objective function and the full set of inequalities.

(3)

(b) Copy and complete the following initial tableau for the simplex method when applied to this problem.

P	x	y	z	s	t	u	
1 0 . . .	–20 2 . . .	–10 1 . . .	–30 2 . . .	0 1 . . .	0 0 . . .	0 0 . . .	0 110 . . .

(2)

(c) Perform one iteration of the simplex method by increasing x.

(5)

(d) Perform a second iteration of the simplex method.

(3)

(e) State how many stools, armchairs and settees should be made in order to maximise the company's profits. Comment on the practicality of this solution.

(3)
(Total 16 marks)

Q31.

When using the simplex method to solve a particular linear programming problem involving three variables, x, y and z the initial tableau is as shown below:

P	x	y	z	s	t	u	
1	–3	–2	–2	0	0	0	0
0	1	1	1	1	0	0	80

0	2	2	1	0	1	0	150
0	2	3	3	0	0	1	180

(a) Express the objective function, P, in terms of x, y and z.

(1)

(b) (i) Apply one iteration of the simplex method by increasing x.

(5)

(ii) Explain how you know that the optimal point has not been reached.

(1)

(c) Perform a second iteration of the simplex method.

(4)

(d) State the maximum possible value of the objective function and the values of x, y and z for which this maximum is reached.

(2)

(e) Write down an inequality in x, y and z which is part of the original problem and which still has slack at the optimal point.

(2)

(Total 15 marks)

Q32.

The simplex method has been applied to a linear programming problem concerning an objective function P in two variables, x and y. The initial tableau T_0 and the tableau T_1, after one iteration of the simplex method, are given by:

T_0

P	x	y	s	t	u	
1	–1	–2	0	0	0	0
0	–1	1	1	0	0	5
0	1	1	0	1	0	15
0	1	–3	0	0	1	3

T_1

P	x	y	s	t	u	
1	0	–5	0	0	1	3
0	0	–2	1	0	1	8
0	0	4	0	1	–1	12
0	1	–3	0	0	1	3

(a) (i) Apply one further iteration of the simplex method to give a new tableau T_2.

(5)

(ii) Explain how you know that the maximum value of P has not yet been reached.

(1)

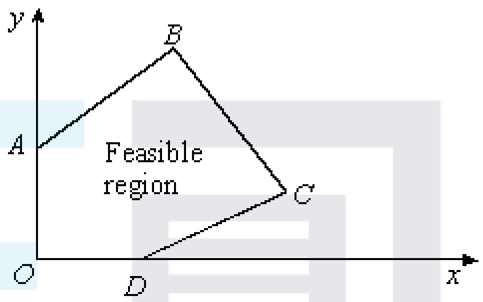
(b) A further iteration of the simplex method leads to tableau T_3.

	P	x	y	s	t	u	
	1	0	0	$\frac{1}{2}$	$1\frac{1}{2}$	0	25
T_3	0	0	0	2	1	1	28
	0	0	1	$\frac{1}{2}$	$\frac{1}{2}$	0	10
	0	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	5

(i) State the maximum value of P and the values of x and y for which this maximum is reached.

(2)

(ii) The figure shows a sketch of the feasible region of the linear programming problem.



For each of the tableaux T_0, T_1, T_2 and T_3 state which of the points O, A, B, C or D it represents.

(3)

(iii) Explain how the original linear programming problem could have been solved by the simplex method with fewer than 3 iterations.

(2)

(Total 13 marks)

Q33.

A linear programming problem consists of maximising an objective function P involving three variables x, y and z. Slack variables s, t, u and v are introduced and the simplex method is applied to solve the problem. Several iterations of the method lead to the following tableau:

P	x	y	z	s	t	u	v	
1	0	0	–4	2	1	0	0	43
0	1	0	–4	1	–1	0	0	17
0	0	1	2	0	2	0	0	26
0	0	0	4	–1	1	1	0	36
0	0	0	1	1	2	1	1	10

(a) Apply one further iteration of the simplex method.

(5)

(b) Explain why your new tableau solves the original problem. State the maximum value of P and the values of x, y and z for which that maximum is reached.

(3)

(c) Apart from $x \geq 0$, $y \geq 0$ and $z \geq 0$, state how many inequalities were involved in the original problem. At the optimum point found in part (b), how many of those inequalities still had some slack?

(2)
(Total 10 marks)

Q34.

When using the simplex method to solve a particular linear programming problem involving two variables x and y the initial tableau was:

P	x	y	s	t	u	
1	–2	–3	0	0	0	0
0	1	2	1	0	0	20
0	1	1	0	1	0	11
0	4	1	0	0	1	23

(a) State the three non-trivial inequalities in x and y and state the objective function.

(3)

(b) Apply one iteration of the simplex method by increasing y. What point does the new tableau represent? Explain how you know that the optimal point has not been reached.

(6)

(c) Perform a second iteration. Interpret this tableau and state the values of the slack variables.

(5)
(Total 14 marks)

Q35.

A factory can produce three items, X-types, Y-types and Z-types. The times needed, in hours, for each item at the three manufacturing stages are shown in the table below. The table also shows the total number of hours available each week for each of the manufacturing stages.

	Manufacturing stage 1	Manufacturing stage 2	Manufacturing stage 3
X–type	1	1	0
Y–type	12	2	3
Z–type	4	1	2
Total hours available	100	40	30

Let x be the number of X-types produced each week, y the number of Y-types and z the number of Z-types.

(a) Write down **three** inequalities imposed on these variables by the availability of the three manufacturing stages.

(3)

(b) Each X-type makes the factory £10 profit, and each Y-type and Z-type makes the factory £40 profit. The factory wishes to maximise its profit. Write down the objective function, P, to be maximised.

(1)

(c) The factory uses the simplex method to solve the problem. At one stage in the solution the tableau is as shown:

P	x	y	z	s	t	u	
1	0	–20	–30	0	1	0	400
0	0	10	3	1	–1	0	60
0	1	2	1	0	1	0	40
0	0	3	2	0	0	1	30

(i) What are the values of the slack variables at this point?

(2)

(ii) Explain how you know that the optimal solution has not yet been reached.

(1)

(iii) Apply one iteration of the simplex method to the tableau, using z as the pivoting variable.

(4)

(iv) Give the complete solution of this linear programming problem.

(3)

(Total 14 marks)

Q36.

The simplex method is to be used to solve the following linear programming problem.

Maximise $9x + 10y$

subject to
$x + y \leq 20$
$3x + 2y \leq 48$
$x \geq 0$
$y \geq 0.$

(a) Use slack variables s and t to convert the first and second inequalities respectively to equalities.

(2)

(b) Starting by increasing x, perform one iteration of the simplex method.

(3)

Two further iterations of the method lead to the equations:

$$P = 200 - 10s - x$$
$$x + y + s = 20$$
$$x - 2s + t = 8.$$

(c) (i) Explain why these equations show that the solution to the problem has been reached.

(ii) Give the solution to the problem.

(iii) Give the number of iterations which would have been needed if y had been increased in the first iteration.

(4)
(Total 9 marks)

Q37.

The following linear programming problem is to be solved using the simplex method.

Maximise $P = x + y$

subject to $4x + 5y \leq 20$

$3x + 2y \leq 12$

$x \geq 0$

$y \geq 0.$

(a) Using slack variables, write the first two constraints as equalities.

(2)

(b) Perform one iteration of the simplex method, starting by increasing x.

(4)

(c) Interpret the results of your iteration in part (b) stating the values of the slack variables and explaining the significance of these values.

(3)
(Total 9 marks)

Q38.

The simplex method is to be used to solve the following linear programming problem:

Maximise $P = 300x + 200y$

subject to $3x + y \leq 3000$ (resource 1)

$6x + 5y \leq 12000$ (resource 2)

$x \geq 0$

$y \geq 0$

(a) Introduce slack variables to convert the resource constraints to equalities.

(2)

(b) Perform two iterations of the simplex algorithm to show that the maximum possible value of P is 500 000.

(6)

(c) Give the values of x and y which achieve the maximum value of P.

(2)

(d) Interpret the values of the slack variables at each stage of your solution.

(2)

(Total 12 marks)

Q39.

The simplex method is to be used to solve the following linear programming problem:

Maximise $9x + 10y$

subject to $x + y \leq 20$

$4x + 5y \leq 94$

$x \geq 0$

$y \geq 0.$

(a) Use slack variables s and t to convert the first two inequalities into equalities.

(2)

(b) (i) Perform one iteration of the simplex method to increase the value of y.

(3)

(ii) Perform a second iteration of the simplex method to complete the solution.

(5)

(Total 10 marks)

Q40.

(a) Solve the following linear programming problem graphically:

Maximise $P = x + 2y$

Subject to $3x + 17y \leq 170$

$7x + 8y \leq 175.$

(5)

(b) Starting from the initial feasible solution $x = y = 0$, apply one iteration of the simplex method to the linear programming problem given in part (a). Identify the point on your graph in part (a) corresponding to the result of your iteration of the simplex method.

(5)

(c) Give the values of the slack variables in the **final** simplex solution to the problem.

(1)

(Total 11 marks)

Q41.

A company producing dining chairs and tables requires a production plan for the next month. The company has £10 000 budgeted to buy materials. Chairs each require 20 of materials and tables £100. Tables each need 15 hours of work from craftsmen and chairs each need 4 hours. There are 1950 hours of craftsman time available per month. The company sells chairs at £80 each and tables at £350. A production plan is required which maximises potential income.

(a) Formulate the problem as a linear programming problem.

(3)

(b) Use a graphical method to solve the problem.

(6)

(c) Had the simplex method been used, slack variables would have been needed.

State what those slack variables would have represented and give their values at the solution.

(2)

(d) Comment on any practical difficulty there may be with your solution to this problem.

(1)

(Total 12 marks)